UHUHWAHHO RAMBOFHENI

Exercise 4

SQL fundamentals

Questions

① SQL Joins.

1. INNER JOIN

```
SELECT
        Employee_Id,
        First_Name,
        Last_Name,
        Department,
        Salary,
        Project_id,
        Project_name,
        ~~Project~~ Budget,
        Status
FROM employees AS A
Inner JOIN Projects AS B
ON A.Employee ID = B.employee.ID
```

Employee_ID
1
2
3
4
5
6

② LEFT JOIN

```
SELECT Employee_id
        first_name,
        Department,
        Salary,
        City,
        Age,
```

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FROM employees AS A
~~Left~~ Jow Project AS B
ON A.employee_id = B.employee_id

Employee_id

employee_id	First_name	Last_name	Department	Salary	City ~~Age~~	Age
1	John	Doe	IT	70 000	New York	35
2	Alice	Smith	HR	60 000	Los Angeles	40
3	Bob	Johnson	finance	75000	Toronto	45
4	David	Brown	IT	72000	London	30
5	Emma	Wilson	Sales	65000	Sydney	28
6	Michael	Clark	finance	80000	New York	50

③ Right JOIN

SELECT
Employee_id,
Project_id,
Projectname,
Budget,
StartDate,
status
FROM projects AS A
RightJoIN employees AS B
ON A.~~Project~~employee_id = B.employee_id

~~Projects~~

Employee_id

Project id	EmployeeID	Project_name	Budget	Start Date	Status
101	1	AI Development	100000	2024-01-10	completed
102	2	Employee Training	50000	2024-02-15	Ongoing
103	3	Cybersecurity Audit	75000	2024-03-05	Pending
104	4	Financial Analysis	90000	2024-04-12	Ongoing
105	5	Market Expansion	65000	2024-05-20	Completed
106	6	Risk Management	80000	2024-06-10	Pending

@FULL OUTER JOIN

```
SELECT Employee_id,
       first_name,
       Last_name,
       Department,
       Salary,
       City,
       Age,
       Project_id,
       Project_name,
       Budget,
       Start_date,
       Status
FROM employee AS A
FULL OUTER JOIN AS B
ON A.Project_id = B.Project_ID
```

Projects.

EmployeeID	ProjectID	firstname	Lastname	Department	Salary	Project name	Budget	Status
1	101	John	Doe	IT	70000	AI develop	100 000	Completed
2	102	Alice	Smith	HR	60000	Employee Tr	50000	ongoing
3	103	Bob	Johnson	finance	75000	CyberAudit	75000	Pending
4	104	David	Brown	IT	72000	Final Analysis	90000	ongoing
5	105	Emma	Wilson	sales	65000	Market expansion	65000	Completed
6	106	Michael	Clark	finance		Risk management	50000	Pending

(4) Aggregate Functions with GROUP BY & HAVING

(13) Total Salary per department, DESC

```
SELECT SUM(DISTINCT Salary) AS Total_Salary,
    Department,

FROM employees
GROUP BY Salary DESC
```

Employees

Department	Total_Salary
IT FINANCE	155 000
IT	142000
Sales	65 000
HR	60 000

~~(14) Salary in Desc~~

(15) Number of employees per department

```
SELECT COUNT(employees) AS employee_count,
        department
FROM employees
GROUP BY department
HAVING >1 'employees'
```

Employees

Employee_count	department
5	IT
2	HR
9	FINANCE
5	Sales

(16) Number of projects per status

```
SELECT COUNT(DISTINCT Project id) A Project count,
        Status
FROM Projects
GROUP BY Status
HAVING =2 projects
```

Projects

Project_count	Status
102	ongoing
104	ongoing
105	completed
106	Pending

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(17) Total project budget per employee

```
SELECT SUM(Budget) AS Total_Project_Budget,
        Employee_id,
        first_name,
        Last_Name.
FROM Projects
GROUP BY ~~employee~~ employee > 150 000 'Budget'
```

Projects:

Employee_id	first_name	last_name	Total_Project_Budget
NULL	NULL	NULL	NULL

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